# SEÁN Ó MURCHÚ



## Senior Software Engineer


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Backend engineer with eight years of experience designing distributed, event-driven systems. Specialises in functional programming, observability, and developer experience.

## EXPERIENCE

### Senior Software Engineer

**Acme Corp**

Jun 2022 – Present Dublin, Ireland

*Platform team lead. Owns the event-sourcing stack underpinning the wider product surface.*

- Led the migration of a customer-facing monolith into a set of event-driven microservices, halving p99 latency.
- Designed and rolled out an event-sourcing platform now used by four product teams.
- Mentored three engineers from mid to senior level.

### Software Engineer

**Liffey Labs**

Jun 2019 – Jun 2021 Remote

- Built and shipped the first version of a SaaS product from scratch alongside a two-person team.
- Introduced the CI/CD pipeline and developer-onboarding flow that scaled the engineering org from 3 to 15.

## VOLUNTEER

### Mentor

**CoderDojo Dublin**

Jun 2020 – Present

*Volunteer-led coding club introducing young people to programming.*

- Weekly mentoring sessions for 10–14 year-olds learning to code.
- Curate the Scala / functional-programming track.

## PROJECTS

### Application

*Hyperion*

Acme Corp

*Distributed task scheduler in Rust*

Lead Architect

Jan 2024 – Aug 2024

- Handled 10k tasks/s on a single node.
- Designed an idempotent retry protocol.

Rust Tokio gRPC

## AREAS OF FOCUS

- Distributed systems: Kafka, CDC, idempotent consumers, event sourcing.
- Functional programming and type-driven design in Scala and Haskell.
- Developer experience: tooling, CI/CD, observability.
- Engineering culture: ADRs, RFCs, mentoring.

## SKILLS

**Languages** - Scala Haskell Python Go Rust

**Infra** - Kafka AWS Terraform Docker Kubernetes

## LANGUAGES

English ●●●●●●

Irish ●●●◐○○

French ●●●●○○

Portuguese ●●◐○○○

## EDUCATION

### M.Sc. in Computer Science

**Tallaght Institute of Technology**

2017 – 2019

Distinction Distributed Systems Type Theory Concurrency

### Computer Science

**Trinity College Dublin**

2014 – 2017

### *Crucible*

*Migration tool for legacy databases*

Maintainer

Sep 2024 – Present

- Schema diffing across PostgreSQL and MySQL.

**Library**

### *Tinkerbell*

*Tiny IRC bot*

Lua

**Other**

### *Quill*

## PUBLICATIONS

### Articles

- *Event Sourcing in Practice*.
  Personal Blog
  14 Jun 2024
  A walk-through of idempotent Kafka consumers and the replay strategy that ships with them.
- *Untyped Note*.
  Jan 2024

### Conference Papers

- *Idempotent Kafka Consumers at Scale*.
  EuroSys
  Apr 2024

### Talks

- *Designing for Failure*.
  ScalaIO
  Sep 2023

### Books

- *Functional Domain Modelling*.
  Self-published
  2022

## INTERESTS

Music - Trad Jazz

Sport - Hurling Climbing

Travel - Japan Iceland

## CERTIFICATIONS

### *Certified Kubernetes Administrator*

Aug 2023 **CNCF**

### *Certified Kubernetes Application Developer*

Jan 2024 **CNCF**

### *AWS Solutions Architect — Professional*

May 2023 **AWS**

### *AWS DevOps Engineer — Professional*

Sep 2024 **AWS**

### *Hashicorp Terraform Associate*

Dec 2022 **Hashicorp**

### *Certified Kubernetes Security Specialist*

Mar 2024 **Linux Foundation**

### *Google Cloud Professional Architect*

Nov 2023 **Google Cloud**

## AWARDS

### *Best Paper Award*

**ScalaConf**

Sep 2024

For *Idempotent Kafka Consumers*.

### *Dean's List*

## REFERENCES

Dr Ada Lovelace

*Seán is a calm, methodical engineer who left every system he touched in a better state than he found it. Recommended without reservation.*

*Pragmatic, curious, unfailingly kind — would hire again in a heartbeat.*